Fiscal Year:	FY 2022	**Task Last Updated:**	FY 12/07/2021
PI Name:	Nicholson, Wayne Ph.D.		
Project Title:	Alteration of Bacillus subtilis DNA Architecture in Space: Global Effects on DNA Supercoiling, Methylation, and the Transcriptome		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	32953	**Congressional District:**	3
Comments:	NOTE: Congressional District is for University of Florida (32611-0001)--Ed., 8/17/2012; changed to 3 (Ed., 7/30/13)		
Project Type:	Flight	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG2. App D: Flight and Ground Space Biology Research
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	No changes to report.		
COI Name (Institution):	Fajardo-Cavazos, Patricia Ph.D. (University of Florida, Gainesville) Wang, Xindan Ph.D. (Indiana University)		
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Task Description:	1) Description of the research product(s): Studying the response of microbial cells to the spaceflight environment is a high-priority issue in Space Biology, due to the importance of microbes in maintaining the health of astronauts and their habitats. Previous International Space Station (ISS) flight experiments (missions Biological Research in Canisters (BRIC)-18, -21, and -23)) using the Gram-positive model bacterium Bacillus subtilis (B. subtilis) have uncovered global changes in the transcriptome, altered mutagenic spectrum of DNA, and significant differences in cell susceptibility to fluoroquinolone antibiotics in spaceflight vs. ground controls. Fluoroquinolones target the enzymes DNA gyrase and topoisomerase IV, which are critical for maintaining DNA supercoiling as well as nucleoid compaction and conformation. Changes in DNA architecture affect many downstream genetic events such as: epigenetic DNA modifications (i.e., the methylome); the location of mutagenic hotspots in DNA; and global gene transcription patterns (i.e., the transcriptome). In turn, such alterations in global gene expression may lead to phenotypic changes observed in spaceflight. Analyses of the data from our prior missions has led us to formulate the following working hypothesis that can be tested experimentally using the BRIC-B. subtilis experimental system: Hypothesis: Exposure to spaceflight microgravity leads to changes in nucleoid architecture, DNA supercoiling, methylation, global transcription patterns, and resulting phenotypes. 2) Type of investigation: ISS with ground controls. 3) Specific aims: Using the B. subtilis model system, compare global: (i) 3-dimensional nucleoid architecture, (ii) DNA supercoiling, (iii) DNA methylation, and (iv) transcriptomes in spaceflight vs. matched ground controls. 4) Experimental plan: Experiments will utilize BRIC-Petri Dish Fixation Unit (PDFU) hardware modified with newly designed inserts mitigating the confounding effect of microgravity on liquid geometry. Four global parameters will be measured: (i) 3-dimensional nucleoid architecture using chromosome conformation capture (Hi-C); (ii) DNA supercoiling using 2-dimensional chloroquine-agarose gel electrophoresis; (iii) DNA methylation using Methyl-seq; and (iv) transcriptomes using RNA-seq. Results will probe at the molecular level the effect of spaceflight microgravity on four fundamental biological processes and their interactions. 5) NASA Relevance: This proposal directly addresses the Microbiology and the Cell and Molecular Biology Elements of the ROSBio-2018 Omnibus announcement and the NASA Strategic Plan Goals, especially Objective 1.2. The proposal directly addresses NASA's Space Biology Science Plan for 2016-2025, most particularly the overarching Microbiology Element Goal "to determine the effect of spaceflight on microbial life, processes, and community dynamics," specifically: 1) MB-1: What underlying genetic, molecular and biochemical processes are influenced by the space environment? (P2); 2) MB-2: What systems biology mechanisms and pathways for microbial life are responsive to spaceflight conditions? (P1, P2); 3) MB-4: What are the mechanisms that effect changes such as the altered virulence or altered drug resistance observed in some organisms during spaceflight? (P1, P2, P3); and 4) MB-5: How does the spaceflight environment influence microbial reproduction, growth, and physiology? (P1, P2).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The goal of the project is to understand at a fundamental (molecular) level how bacteria sense and respond to the spaceflight environment. At present it is known that the phenotypic behavior of microbes is altered when confronted with spaceflight microgravity, but to date the underlying mechanism(s) involved have remained obscure. The project is to measure four fundamental molecular processes in cells of the model bacterium Bacillus subtilis: three-dimensional nucleoid architecture, DNA supercoiling, DNA methylation, and transcriptomes. These processes will be measured and compared in spaceflight (FL) vs. matched ground control (GC) samples. Deeper understanding of the molecular responses of bacteria to prolonged spaceflight has important implications for understanding disease processes and performance of microbes in bioreactors both in space and on Earth.
	The Specific Aims of this project are: Using the B. subtilis model system in spaceflight (FL) vs. matched ground controls (GC), compare global: A. 3-dimensional nucleoid architecture, B. DNA supercoiling, C. DNA methylation, and D. transcriptomes. We have been informed that our spaceflight experiment to the ISS is designated BRIC-26, which is tentatively scheduled for launch on SpaceX CRS-25 in late May 2022. Despite delays imposed by COVID-19, we have been performing pre-flight testing of the spaceflight hardware to be used (i.e., modified Petri dish insert), and we have also been optimizing protocols to accomplish Specific Aims A and B. Testing of new Flight Hardware. In two previous spaceflight experiments to the ISS (BRIC-21 and BRIC-23), we noted that the transcriptome of B. subtilis differed substantially and reproducibly between matched FL and GC samples, and that liquid in BRIC-PDFUs assumed a different geometry in FL (toroid) vs. GC (flat) (Ed. Note. Reference: Morrison MD, Fajardo-Cavazos P, Nicholson WL. "Comparison of Bacillus subtilis transcriptome profiles from two separate missions to the International Space Station." npj Microgravity 2019 Jan;5:1. https:// ; PMID: 30623021; PMCID: PMC6323116). The observations suggested that differences in FL vs. GC gene expression might be due to different liquid geometries and not spaceflight microgravity per se. To address this issue, we designed, built, and tested new Petri dish inserts intended to keep liquid in the bottom of the dish regardless of gravity or its absence. The new dishes proved to function as designed in 2-second drop tower tests and were verified to be biocompatible with B. subtilis. In preliminary experiments we have shown that under simulated "ISS" growth conditions, a culture grown in a single PDFU containing a modified Petri dish insert yields a final population of ~10^9 cells, sufficient for all downstream experiments. Details can be found in an article due to be published in late December 2021. (Ed. Note. Reference: Nicholson WL, Fajardo-Cavazos P, Turner C, Currie TM, Gregory G, Jurca T, Weislogel M. "Design and validation of a device for mitigating fluid microgravity effects in Biological Research in Canister (BRIC) spaceflight hardware." Front. Space Technol. 2:797518. In review December 2021. doi:10.3389/frspt.2021.797518). Specific Aim A. 3-dimensional nucleoid architecture. With this aim we propose to measure the three-dimensional nucleoid architecture of B. subtilis cells grown either in FL or GC samples using a high-definition chromosome capture (Hi-C) technique. To our knowledge, this has never been attempted before on bacterial cells cultivated in BRIC-PDFU canisters in spaceflight. We are currently working out the optimum conditions in collaboration with the laboratory of Dr. Xindan Wang at Indiana University, who has expertise in Hi-C investigations of B. subtilis chromosome architecture.

Task Progress:	In this procedure, cells are cultivated in Dual-Chamber BRIC-PDFU hardware as determined in prior BRIC-21 and BRIC-23 experiments (i.e., TSYG liquid medium, static culture, 25 hours, ISS ambient temperature of 22-23°C). The medium is first injected to initiate spore germination and growth. At 25 hours, formaldehyde is injected to crosslink the DNA in the cells. The liquids are mixed and incubated at ISS-ambient temperature for a pre-set interval, then frozen in the onboard -80°C MELFI freezer and subsequently returned in the frozen state to Earth. Upon return to the lab, samples are thawed and processed for subsequent Hi-C mapping. We have been optimizing several parameters: concentration of formaldehyde, duration of fixation, quenching, and cell recovery conditions. Each series of optimization experiments requires running the samples through the entire Hi-C process and visualization of the results, which can take up to a month. At present, we have completed 4 iterations of optimization. The best results we have obtained so far are: Equal volume of 20% formaldehyde (10% final concentration); mix; 2 hours fixation at ISS-ambient temperature (22-23°C); freeze at -80°C (MELFI for FL; lab freezer for GC); thaw in lab, quench with 0.4M glycine, 10 min; centrifuge and wash cells twice with ice-cold PBS; adjust final cell density and freeze; ship on dry ice to Wang lab for Hi-C determination. The results indicate that Hi-C data can be obtained from cells cultivated under "ISS" conditions. The data indicate that under "ISS" growth conditions, fewer short- and long-range chromosome interactions are apparent than when cells are grown under standard laboratory conditions. This is likely due to the lower temperature and limited oxygen in PDFUs, hence slower growth rate, of cells growing under ISS-ambient conditions. Specific Aim B. DNA supercoiling of proxy plasmid pUB110. In this Aim we propose to measure the supercoiling state of DNA in FL vs. GC samples using the small, covalently closed circular plasmid pUB110 carried by B. subtilis strain WN112 as a proxy for chromosomal DNA supercoiling. To date we have cultivated strain WN112 under "ISS" conditions and obtained sufficient plasmid DNA from each single sample to perform 4-6 supercoiling determinations. Plasmid DNA from strain WN112 was isolated using a commercial kit and electrophoresed through 1.4% agarose gels containing the intercalating agent chloroquine at final concentrations of 0, 2, or 4 micrograms/mL. It is evident from this series of gels that a chloroquine concentration of 4 micrograms/mL more efficiently resolves the topoisomers of interest. However, the goal is to accurately measure the average numbers of supercoils in FL vs. GC samples; a slightly higher concentration of chloroquine (5 or 6 micrograms/mL) may be required for this. At present, we have shown that we can separate topoisomers from cells grown under "ISS conditions" in the lab. We desire to calculate absolute numbers of superhelical turns in plasmid pUB110; for this we will construct standard curves by relaxation of purified plasmid DNA with Topoisomerase I in the presence of increasing concentrations of ethidium bromide, as previously described (Ed. Note. Reference: Nicholson WL, Setlow P. "Dramatic increase in negative superhelicity of plasmid DNA in the forespore compartment of sporulating cells of Bacillus subtilis." J Bacteriol 1990 Jan;172(1):7-14. https:// ; PMID: 2104613; PMCID: PMC208394). Specific Aims C and D. We have previously measured transcriptomes in space during the BRIC-21 and -23 experiments, and do not anticipate any problems. We expect to be able to isolate sufficient DNA from samples to measure methylomes using PacBio sequencing. For verification purposes, we will grow cells under simulated "ISS conditions" and perform DNA and RNA extractions, and will quantify nucleic acid yields and quality in order to assure that we will be able to obtain sufficient material for post-flight analyses.
Bibliography Type:	Description: (Last Updated: 07/25/2025)
Abstracts for Journals and Proceedings	Nicholson WL, Fajardo-Cavazos P, Turner C, Jurca T, Weislogel M. "Design and testing of a new insert to mitigate microgravity effects on liquid cultures in BRIC-PDFU or BRIC-LED hardware." 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. Abstracts. 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020; Abstract: 112. , Nov-2020
Abstracts for Journals and Proceedings	Nicholson WL, Fajardo-Cavazos P, Currie TM, Jurca T, Weislogel M. "Biocompatibility testing of a new BRIC hardware insert designed to mitigate microgravity effects on liquid cultures in spaceflight." 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. Abstracts. 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021.Abstract: 2021102. , Nov-2021
Articles in Peer-reviewed Journals	Fajardo-Cavazos P, Nicholson WL. "Mechanotransduction in prokaryotes: A possible mechanism of spaceflight adaptation." Life (Basel). 2021 Jan 7;11(1):E33. https://doi.org/10.3390/life11010033 ; PMID: 33430182; PMCID: PMC7825584 , Jan-2021
Articles in Peer-reviewed Journals	Nicholson WL, Fajardo-Cavazos P, Turner C, Currie TM, Gregory G, Jurca T, Weislogel M. "Design and validation of a device for mitigating fluid microgravity effects in Biological Research in Canister (BRIC) spaceflight hardware." Front. Space Technol. 2021 Dec 22;2:797518. https://doi.org/10.3389/frspt.2021.797518 , Dec-2021